



Statistics of Financial Markets

Seminar 9: ARCH and GARCH models



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material

- ▣ AI: Artificial Intelligence
- ▣ AIC: Akaike Information Criterion
- ▣ ARCH: AutoRegressive Conditional Heteroskedasticity
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BIC: Bayesian Information Criterion
- ▣ BVB: Bursa de Valori Bucureşti (the Bucharest Stock Exchange)
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ DM: Diebold–Mariano (test)
- ▣ EGARCH: Exponential GARCH
- ▣ EODHD: EOD Historical Data (market-data provider)
- ▣ EWMA: Exponentially Weighted Moving Average
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GJR: Glosten–Jagannathan–Runkle (asymmetric GARCH)
- ▣ GJR-GARCH: Glosten–Jagannathan–Runkle GARCH
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ IGARCH: Integrated GARCH
- ▣ LR: Likelihood Ratio (test)
- ▣ MLE: Maximum Likelihood Estimation
- ▣ QLIKE: Quasi-LIKelihood (loss function)
- ▣ SE: Standard Error

Today's Question and Route

- **Question:** how can we model, estimate and forecast a volatility that changes every day?
 - ▶ this seminar comes **before** Lecture 9: the section “Prerequisites for Today” gives every definition the tasks use
- Route
 - ▶ Part A: GARCH algebra, forecasts and VaR, the ARCH likelihood, asymmetric news impact, on paper
 - ▶ Part B: GARCH estimation, asymmetry and out-of-sample forecasts on the S&P 500, DAX, BET, Bitcoin and BVB stocks, each with an interpretation question
 - ▶ Part C: an open question for a project and an AI answer to audit
- Notebook for today: [open the seminar notebook in Google Colab](#); each task names its notebook section
- The seminar is for practice and is not graded; the solutions of [Proposed] tasks are discussed in class

Exercise Map

Task	Question	Type	Model
A1, A2	persistence, long-run variance, half-life and one GARCH(1,1) step	Solved, Proposed	A1
A3, A4	multi-step forecasts, 10-day volatility and VaR 1%	Solved, Proposed	A3
A5, A6	the ARCH(1) likelihood and kurtosis; news impact in GJR-GARCH and EGARCH	Solved, Proposed	A5
B1, B2	GARCH(1,1)-t estimates: DAX; BET, Banca Transilvania, Bitcoin	Solved, Proposed	B1
B3, B4	asymmetry and the leverage effect: S&P 500; BET, Bitcoin, TLV	Solved, Proposed	B3
B5, B6	GARCH against EWMA out of sample, VaR 1% exceedances: S&P 500; DAX, BET	Solved, Proposed	B5
C1, C2	is Bitcoin volatility becoming more like equity volatility? what is wrong in an AI answer?	Proposed	B1, B3

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model

Data Used

Series	Source	Price used	Period
S&P 500, DAX, BET	EODHD	close	2000–2026
Bitcoin	EODHD	close, 7 days a week	2014–2026
Banca Transilvania (TLV)	EODHD	adjusted close	2010–2026

- Daily log returns in %: $r_t = 100 (\ln P_t - \ln P_{t-1})$; weekends and repeated holiday closes are dropped (except Bitcoin); the last day is 18 September 2026
- TLV: the days 30 and 31 May 2016 are removed (a price adjustment applied one day late, [Chapter 2](#))
- In the notebook: `returns('dax')`, `fit(r, dist='t')` (package `arch`); no account or key is needed

Prerequisites for Today (1/5): Conditional Variance, ARCH and GARCH

- $r_t = \mu + \varepsilon_t$, $\varepsilon_t = \sigma_t z_t$; z_t i.i.d., mean 0, variance 1 (the **innovations**)
 - ▶ $\sigma_t^2 = \text{Var}(r_t \mid \text{past})$: the **conditional variance**, known one day ahead
 - ▶ the shocks ε_t are uncorrelated, but their squares are correlated (volatility clustering, [Chapter 8](#))
- **ARCH(1)** [Engle \(1982\)](#): $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2$, $\omega > 0$, $0 \leq \alpha < 1$
 - ▶ in words: today's variance = a constant ω plus a share α of yesterday's squared shock
 - ▶ unconditional (average) variance $\omega/(1 - \alpha)$
 - ▶ with Normal z_t and $3\alpha^2 < 1$: kurtosis $3(1 - \alpha^2)/(1 - 3\alpha^2) > 3$
- **GARCH(1,1)** [Bollerslev \(1986\)](#): $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$, $\omega > 0$, $\alpha, \beta \geq 0$
 - ▶ ω : a constant that sets the long-run level; α : reaction to news; β : memory (weight of yesterday's variance)
 - ▶ $\alpha + \beta < 1$: the model is stationary
 - ▶ **EWMA** ([Chapter 8](#)): $\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) r_{t-1}^2$, a GARCH with $\omega = 0$, $\alpha + \beta = 1$ (**IGARCH**)

Prerequisites for Today (2/5): Persistence and Forecasts

- ▣ **Persistence** $\alpha + \beta$; **long-run variance** $\bar{\sigma}^2 = \omega / (1 - \alpha - \beta)$; annual volatility $\sqrt{252 \bar{\sigma}^2}$
 - ▶ **half-life**: $h_{1/2} = \ln 0.5 / \ln(\alpha + \beta)$ days, the time for half of a variance shock to disappear
- ▣ Forecasts made at the end of day t
 - ▶ E_t : the expectation given the data up to day t ; h : the horizon in days; H : the length of the holding period
 - ▶ one day: $\sigma_{t+1}^2 = \omega + \alpha(r_t - \mu)^2 + \beta\sigma_t^2$
 - ▶ h days: $E_t[\sigma_{t+h}^2] = \bar{\sigma}^2 + (\alpha + \beta)^{h-1}(\sigma_{t+1}^2 - \bar{\sigma}^2)$
 - ▶ H -day variance: $\sum_{h=1}^H E_t[\sigma_{t+h}^2] = H\bar{\sigma}^2 + (\sigma_{t+1}^2 - \bar{\sigma}^2) \frac{1 - (\alpha + \beta)^H}{1 - \alpha - \beta}$
- ▣ **VaR 1%** (value at risk, [Chapter 6](#)): $\text{VaR}_{t+1} = -(\mu + \sigma_{t+1} q_{0.01}(z))$
 - ▶ $q_{0.01}(z)$: the 1% quantile of the innovation z ; the VaR is a positive loss in %
 - ▶ Normal: $q_{0.01} = -2.326$; standardised Student-t: $q_{0.01} = t_\nu^{-1}(0.01) \sqrt{(\nu - 2)/\nu}$
 - ▶ $t_\nu^{-1}(0.01)$: the 1% quantile of the Student-t with ν degrees of freedom, for example $t_5^{-1}(0.01) = -3.365$

Prerequisites for Today (3/5): Maximum Likelihood

- ▣ Conditional Gaussian log-likelihood, with σ_t^2 computed recursively:
 - ▶ $\ell(\theta) = \sum_{t=2}^T \ell_t, \quad \ell_t = -\frac{1}{2} \left[\ln(2\pi) + \ln \sigma_t^2 + \frac{(r_t - \mu)^2}{\sigma_t^2} \right]$
 - ▶ $\theta = (\mu, \omega, \alpha, \beta)$: the parameters; T : the number of returns; ℓ_t : the contribution of day t
 - ▶ **MLE**: $\hat{\theta}$ maximises ℓ ; numerically, with $\omega > 0, \alpha, \beta \geq 0, \alpha + \beta < 1$
- ▣ Standard errors (SE)
 - ▶ classic: from the curvature of ℓ at its maximum
 - ▶ robust [Bollerslev & Wooldridge \(1992\)](#): valid when z_t is not Normal; arch reports the robust ones
- ▣ Heavy-tailed innovations ([Chapter 2](#)): standardised Student-t with ν degrees of freedom [Bollerslev \(1987\)](#); skewed t [Hansen \(1994\)](#)
 - ▶ compare models with the **LR** (likelihood ratio) test $2(\ell_1 - \ell_0) \sim \chi^2(r)$, and with AIC/BIC ([Chapter 6](#))
 - ▶ ℓ_1, ℓ_0 : maximised log-likelihoods of the larger and the smaller model; r : number of restrictions;
 $\chi_{0.95}^2(1) = 3.84$

Prerequisites for Today (4/5): Asymmetry

- ▣ **Leverage effect:** falls raise volatility more than rises of the same size [Christie \(1982\)](#)
- ▣ **GJR-GARCH** [Glosten, Jagannathan & Runkle \(1993\)](#): $\sigma_t^2 = \omega + (\alpha + \gamma I_{t-1})\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$, $I_{t-1} = 1$ if $\varepsilon_{t-1} < 0$
 - ▶ I_{t-1} : indicator of a negative shock yesterday; γ : the extra reaction to bad news
 - ▶ leverage effect: $\gamma > 0$; persistence $\alpha + \beta + \gamma/2$ for symmetric z_t
- ▣ **EGARCH** [Nelson \(1991\)](#): $\ln \sigma_t^2 = \omega + \alpha(|z_{t-1}| - E|z_{t-1}|) + \gamma z_{t-1} + \beta \ln \sigma_{t-1}^2$
 - ▶ models $\ln \sigma_t^2$, so the variance is always positive; $|z_{t-1}| - E|z_{t-1}|$: the size of yesterday's standardised shock above its mean
 - ▶ $E|z| = \sqrt{2/\pi} = 0.7979$ for Normal z ; leverage effect: $\gamma < 0$
- ▣ **News impact curve** [Engle & Ng \(1993\)](#): σ_t^2 as a function of ε_{t-1} , with σ_{t-1}^2 fixed
 - ▶ symmetric parabola for GARCH; steeper on the left for GJR and EGARCH with leverage

Prerequisites for Today (5/5): Checking and Comparing Forecasts

- Standardised residuals $\hat{z}_t = (r_t - \hat{\mu})/\hat{\sigma}_t$: close to i.i.d. if the model is right
 - ▶ Ljung–Box $Q(10)$ on $\hat{z}_t^2$ (Chapter 7) Ljung & Box (1978): no ARCH effect left if not rejected
 - ▶ sign-bias test Engle & Ng (1993): $\chi^2(3)$; a rejection means a missing asymmetry
- **Out of sample**: the forecast h_t for day t uses only data up to $t - 1$; proxy of the true variance: r_t^2
 - ▶ **QLIKE** loss Patton (2011): $L_t = r_t^2/h_t + \ln h_t$; the smaller the mean, the better the forecasts
 - ▶ **DM** (Diebold–Mariano) test Diebold & Mariano (1995): t statistic of the mean of $L_t^A - L_t^B$ with HAC standard errors; $t < -1.96$: A is better
- **VaR exceedance**: a day with $r_t < -\text{VaR}_t$; a correct VaR 1% is exceeded on about 1% of days

A1: GARCH(1,1) Algebra [Solved]

Task

- ▣ Daily returns in %: $\omega = 0.02$, $\alpha = 0.08$, $\beta = 0.90$, $\mu = 0$, 252 trading days per year.
- ▣ 1. Compute the persistence and the long-run variance.
- ▣ 2. Compute the annualised long-run volatility.
- ▣ 3. Compute the half-life of a variance shock.
- ▣ 4. Today $\sigma_t^2 = 1.5$ and $r_t = -3\%$; compute σ_{t+1}^2 and σ_{t+1} .
- ▣ Report: four numbers and one sentence.

Solution

- ▣ 1. $\alpha + \beta = 0.98$; $\bar{\sigma}^2 = 0.02/(1 - 0.98) = 1.000$
- ▣ 2. $\sqrt{252 \times 1.000} = 15.87\%$ per year
- ▣ 3. $\ln 0.5 / \ln 0.98 = -0.69315 / (-0.02020) = 34.3$ days
- ▣ 4. $0.02 + 0.08 \times 9 + 0.90 \times 1.5 = 2.090$; $\sigma_{t+1} = 1.446\%$
- ▣ A fall of 3% raises the variance by 39%; half of the excess is gone after 34.3 trading days, about 7 weeks.

A2: A More Reactive Market [Proposed]

Solution discussed in class

Task

- $\omega = 0.05$, $\alpha = 0.12$, $\beta = 0.85$, $\mu = 0$; today $\sigma_t^2 = 2.0$ and $r_t = +1.5\%$. Model: A1.
- 1. Compute the persistence, the long-run variance and the annualised long-run volatility.
- 2. Compute the half-life.
- 3. Compute σ_{t+1}^2 .
- 4. Write EWMA with $\lambda = 0.94$ as a GARCH(1,1) and say which of the quantities in 1–2 exist for it.
- Report: four numbers and two sentences.

A3: Forecasts and VaR 1% [Solved]

Task

- The model of A1, with $\sigma_{t+1}^2 = 2.090$ and Normal innovations.
- 1. Compute $E_t[\sigma_{t+5}^2]$ and $E_t[\sigma_{t+10}^2]$.
- 2. Compute the 10-day variance and volatility.
- 3. Compare with the square-root-of-time rule $\sqrt{10} \sigma_{t+1}$ and with $\sqrt{10} \bar{\sigma}^2$.
- 4. Compute the one-day and the 10-day VaR 1%.
- Report: six numbers and one sentence.

Solution

- 1. $1 + 0.98^4(2.090 - 1) = 1 + 0.9224 \times 1.09 = 2.005$;
 $1 + 0.98^9 \times 1.09 = 1.909$
- 2. $10 \times 1 + 1.09 \times (1 - 0.98^{10})/0.02 =$
 $10 + 1.09 \times 9.146 = 19.97$; volatility 4.47%
- 3. $\sqrt{10 \times 2.090} = 4.57\%$ (too high),
 $\sqrt{10 \bar{\sigma}^2} = \sqrt{10 \times 1} = 3.16\%$ (too low)
- 4. $2.326 \times 1.446 = 3.36\%$; $2.326 \times 4.47 = 10.40\%$
- The high variance decays slowly: the 10-day risk lies between the two simple rules, closer to the first.

A4: Forecasts with Student-t Innovations [Proposed]

Solution discussed in class

Task

- ▣ The model of A2, with $\sigma_{t+1}^2 = 2.020$ and standardised Student-t innovations with $\nu = 5$ ($t_5^{-1}(0.01) = -3.365$). Model: A3.
- ▣ 1. Compute $E_t[\sigma_{t+10}^2]$ and the 10-day variance.
- ▣ 2. Compute the quantile $q_{0.01}$ of the standardised t.
- ▣ 3. Compute the one-day and the 10-day VaR 1% with t and with Normal innovations.
- ▣ Report: six numbers and one sentence.

A5: The ARCH(1) Likelihood, Step by Step [Solved]

Task

- ▣ ARCH(1) with $\omega = 0.5$, $\alpha = 0.5$, $\mu = 0$, Normal z_t ; observed: $r_1, \dots, r_4 = 0.5; -1.2; 2.0; -0.3$.
- ▣ 1. Compute σ_2^2 , σ_3^2 and σ_4^2 .
- ▣ 2. Compute ℓ_2 , ℓ_3 , ℓ_4 and the conditional log-likelihood.
- ▣ 3. Compute the unconditional variance and the kurtosis of this ARCH(1).
- ▣ Report: seven numbers and one sentence.

Solution

- ▣ 1. $0.5 + 0.5 \times 0.25 = 0.625$; $0.5 + 0.5 \times 1.44 = 1.220$; $0.5 + 0.5 \times 4 = 2.500$
- ▣ 2. $\ell_2 = -\frac{1}{2}[1.8379 + \ln 0.625 + 1.44/0.625] = -1.836$; $\ell_3 = -2.658$; $\ell_4 = -1.395$; sum -5.889
- ▣ 3. $0.5/(1 - 0.5) = 1.0$; $K = 3 \times 0.75/0.25 = 9.0$
- ▣ Day 3 ($r_3 = 2$ after a calm day) contributes least: a large return when σ_t^2 is small is “unlikely”.

A6: Good and Bad News [Proposed]

Solution discussed in class

Task

- GJR-GARCH: $\omega = 0.02$, $\alpha = 0.03$, $\gamma = 0.12$, $\beta = 0.90$; EGARCH: $\omega = 0$, $\alpha = 0.12$, $\gamma = -0.08$, $\beta = 0.98$; in both $\sigma_t^2 = 1$, $\mu = 0$. Model: A1.
- 1. Compute the GJR σ_{t+1}^2 after $\varepsilon_t = -2$ and after $\varepsilon_t = +2$, and their ratio.
- 2. Compute the GJR persistence and half-life.
- 3. Compute the EGARCH σ_{t+1}^2 for the same two shocks (use $E|z| = 0.7979$).
- Report: six numbers and one sentence.

B1: GARCH(1,1)-t for the DAX [Solved]

- ▣ **Question:** how persistent is DAX volatility, and how far is the DAX from its long-run volatility today?
- ▣ **Inputs:** DAX closes since 2000, daily log returns in %
- ▣ **Tasks**
 1. Estimate GARCH(1,1) with Student-t innovations and report the parameters with robust standard errors.
 2. Compute the persistence, the half-life and the annualised long-run volatility, and compare it with the sample volatility.
 3. Report the peaks of the annualised conditional volatility in 2008 and in 2020, and its value on the last day.
 4. Draw the returns since 2018 with $\pm 2\hat{\sigma}_t$ bands.
 5. Interpretation: should a risk manager use the long-run volatility or the current one for tomorrow?
- ▣ **Report:** a table of five parameters, five numbers, the chart and two sentences
- ▣ **Notebook:** section B1

B1: Solution [Solved]



	μ	ω	α	β	ν
estimate	0.081	0.0206	0.097	0.895	7.08
robust SE	0.012	0.0049	0.011	0.012	0.62

- $T = 6\,786$; $\alpha + \beta = 0.992$, half-life $\ln 0.5 / (-0.00784) = 88.5$ days; long-run volatility 25.9% against 22.4% in the sample
- Peaks: 80% (2008), 82% (2020); last day: 13.0%; 5.2% of the days fall outside $\pm 2\hat{\sigma}_t$
- Interpretation: the current one: VaR for tomorrow needs σ_{t+1} ; the long-run level matters only for long horizons, and it is imprecise because $1 - \alpha - \beta = 0.0078$ is small

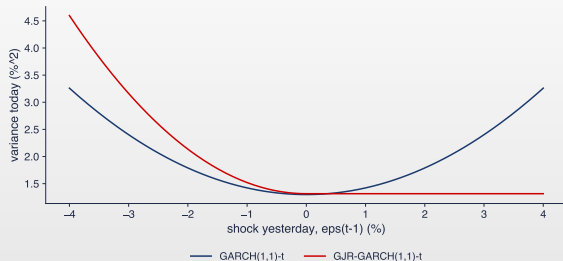
B2: BET, Banca Transilvania and Bitcoin [Proposed]

- **Question:** which of the BET, Banca Transilvania and Bitcoin has the most persistent volatility? Model: B1.
- **Inputs:** BET since 2000, TLV since 2010 (without 30–31 May 2016), Bitcoin since 2014; daily log returns in %
- **Tasks**
 1. Estimate GARCH(1,1)-t for each series and report α , β and ν .
 2. Compute the persistence, the half-life and the long-run volatility where they exist.
 3. Compare the long-run with the sample volatility (annualise Bitcoin with 365 days).
 4. Interpretation: what does an estimate $\hat{\alpha} + \hat{\beta} = 1$ tell a risk manager?
- **Report:** one table and two sentences
- **Notebook:** section B2

B3: The Leverage Effect in the S&P 500 [Solved]

- ▣ **Question:** do falls raise the volatility of the S&P 500 more than rises of the same size?
- ▣ **Inputs:** S&P 500 closes since 2000, daily log returns in %
- ▣ **Tasks**
 1. Estimate GARCH(1,1)-t and GJR-GARCH(1,1)-t; report γ with its robust t statistic.
 2. Compute the LR statistic of GJR against GARCH, its p-value and the two BIC values.
 3. Run the sign-bias test on the standardised residuals of GARCH-t.
 4. Draw both news impact curves and compute the GJR variance after shocks of -2% and $+2\%$.
 5. Interpretation: how does the leverage effect change the VaR on the day after a fall?
- ▣ **Report:** four statistics with p-values, two variances, the chart and two sentences
- ▣ **Notebook:** section B3

B3: Solution [Solved]



- ▣ GJR: $\hat{\alpha} = 0.000$, $\hat{\gamma} = 0.205$ ($t = 10.03$), $\hat{\beta} = 0.881$; LR = 235.7 ($p < 0.001$); BIC 18169.0 $\rightarrow$ 17942.2
- ▣ Sign bias on GARCH-t residuals: $\chi^2(3) = 40.7$ ($p < 0.001$), sign $t = 3.56$
- ▣ GJR variance after -2% : 2.136; after $+2\%$: 1.315; ratio 1.62
- ▣ Interpretation: only falls raise the variance ($\hat{\alpha} = 0$); after a fall the VaR is about $\sqrt{1.62}$ times the VaR after an equal rise; a symmetric GARCH understates risk after bad days

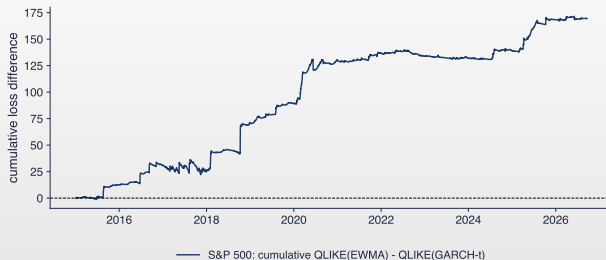
B4: Asymmetry on the BVB and in Bitcoin [Proposed]

- ▣ **Question:** is there a leverage effect in the BET, in Banca Transilvania and in Bitcoin? Model: B3.
- ▣ **Inputs:** BET since 2000, TLV since 2010, Bitcoin since 2014; daily log returns in %
- ▣ **Tasks**
 1. Estimate GJR-GARCH(1,1)-t and EGARCH(1,1)-t; report both γ with robust t statistics.
 2. Compute the LR test of GJR against GARCH and the change in BIC.
 3. Compute the GJR ratio of the variances after -2% and $+2\%$.
 4. Interpretation: would you use an asymmetric model for each of the three? Why?
- ▣ **Report:** one table and two sentences
- ▣ **Notebook:** section B4

B5: GARCH against EWMA for the S&P 500 [Solved]

- **Question:** does GARCH(1,1)-t forecast tomorrow's variance of the S&P 500 better than EWMA, and is its VaR 1% exceeded on 1% of days?
- **Inputs:** S&P 500 since 2000; forecasts for every day since 2 January 2015
- **Tasks**
 1. Re-estimate GARCH(1,1)-t every 250 days on all data up to that day and compute the one-day forecasts h_t ; compute EWMA with $\lambda = 0.94$.
 2. Compute the mean QLIKE of both models and the Diebold–Mariano t statistic.
 3. Count the exceedances of the GARCH-t VaR 1% and of the EWMA-Normal VaR 1% and compare with the expected number.
 4. Draw the cumulative difference of the losses.
 5. Interpretation: which model would you give to a risk manager, and what would you still fix?
- **Report:** four numbers, two counts, the chart and two sentences
- **Notebook:** section B5

B5: Solution [Solved]



- 2944 days; mean QLIKE: GARCH-t 0.761, EWMA 0.819; DM: mean difference -0.058 , $t = -3.69$ ($p < 0.001$): GARCH-t is better
- VaR 1% exceedances: GARCH-t 49 (1.66%), EWMA-Normal 65 (2.21%); expected about 29
- Interpretation: GARCH-t, but both VaR models are exceeded too often; add the leverage effect (B3) and test the exceedances formally (Chapter 10)

B6: GARCH against EWMA for the DAX and the BET [Proposed]

- **Question:** does the result of B5 hold for the DAX and for the BET? Model: B5.
- **Inputs:** DAX and BET since 2000; forecasts for every day since January 2015
- **Tasks**
 1. Compute the out-of-sample GARCH(1,1)-t and EWMA forecasts as in B5.
 2. Compute the mean QLIKE of both models and the DM test.
 3. Count the VaR 1% exceedances of both models.
 4. Interpretation: for which index does GARCH help most, and why?
- **Report:** one table and two sentences
- **Notebook:** section B6

C1: Is Bitcoin Volatility Becoming Equity-Like? [Proposed]

- ▣ **Question:** has Bitcoin volatility moved closer to the volatility of the S&P 500 since 2021?
- ▣ **Inputs:** Bitcoin and S&P 500 daily log returns, 2015–2019 and 2021–2026; models: B1, B3
- ▣ **Tasks**
 1. Estimate GJR-GARCH(1,1)-t for both series in both periods and report the persistence, γ and ν .
 2. Compute the correlation of the logarithms of the two GARCH-t volatilities on common days in each period.
 3. List the events of 2020–2024 that could explain a change.
 4. Interpretation: how would you separate a lasting change from a change caused by one crisis?
- ▣ **Report:** a table and a plan for a project
- ▣ **Notebook:** section C1

C2: Audit an AI Answer [Proposed]

- A student asked an AI assistant to interpret a GARCH(1,1)-t fit of the S&P 500 (daily, since 2000): $\hat{\omega} = 0.0157$, $\hat{\alpha} = 0.123$, $\hat{\beta} = 0.872$, $\hat{\nu} = 6.26$. The answer:
 - (a) $\alpha + \beta = 0.995$, so a volatility shock disappears within about a week.
 - (b) The long-run variance is $\omega/(1 - \alpha) = 0.018$, an annual volatility of 2.1%.
 - (c) A GJR fit gives $\gamma = 0.205 > 0$: good news raises volatility more than bad news.
 - (d) Ljung-Box Q(10) of the squared standardised residuals = 13.6, $p = 0.19$: the GARCH-t model is correct.
 - (e) With Student-t innovations the one-day VaR 1% is 2.326 times $\sigma(t+1)$.
 - (f) EWMA with $\lambda = 0.94$ is a GARCH(1,1) with $\omega = 0$, $\alpha = 0.06$, $\beta = 0.94$.
- Tasks
 - ▶ 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number from the notebook (section C2).
 - ▶ 2. Report: a list of six verdicts with one line of justification each.

What You Should Take from Today

- ▣ $\alpha + \beta$ decides everything long-run: half-life, long-run variance, the shape of the forecasts
- ▣ Multi-day risk is the sum of the daily variance forecasts, not $\sqrt{H}$ times today's volatility
- ▣ Heavy-tailed innovations change the VaR quantile; asymmetry changes the VaR after bad days
- ▣ Compare forecasts out of sample (QLIKE, DM), not only in sample (AIC, BIC)
- ▣ An AI answer is a draft: check the formula of the long-run variance, the sign of γ and the quantile

After the Seminar

- ▣ Lecture 9 develops each topic of today: ARCH and GARCH, estimation, innovations, asymmetry, diagnostics, forecasts, VaR
- ▣ Try the [Proposed] tasks in the notebook; the solutions are discussed in class
- ▣ C1 can grow into a team project: rolling windows, Ethereum and gold as controls
- ▣ Reading: [Franke, Härdle & Hafner \(2019\)](#), Ch. 13; exercises in [Borak, Härdle & López-Cabrera \(2013\)](#), Ch. 13; [Tsay \(2010\)](#), Ch. 3

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